[CANCER RESEARCH 48, 2015–2020, April 15, 1988]

Repair of 4,5′,8-Trimethylpsoralen plus Light-induced DNA Damage in Normal and Fanconi's Anemia Cell Lines[1]

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ABSTRACT

In conditions in which solely monoadducts (MA) are induced in DNA, *i.e.*, treatment with 4,5′,8-trimethylpsoralen and 405 nm radiation, Fanconi's anemia cells (FA) appear to be more sensitive than normal human fibroblasts (1 BR/3) to cytotoxicity. The repair of such induced MA is impaired in FA compared to normal cells. When increasing the proportion of DNA interstrand cross-links (CL) over MA using a reirradiation protocol, the differential sensitivity between FA and normal human cells increases. Moreover, for a constant number of total adducts or at different ratios of CL over MA, the repair of CL is systematically hampered in FA as compared to normal cells. Incision of CL being progressively diminished by increasing amounts of MA in normal cells (D. Papadopoulo, D. Averbeck, and E. Moustacchi, Photochem. Photobiol., *47:* 321–326, 1988), we show here that it is even more so for FA cells.

INTRODUCTION

FA[2] is an autosomal recessive disorder associated with chromosomal instability (1, 2) and predisposition to cancer (3). It is generally accepted that FA cells are more or less specifically hypersensitive to DNA cross-linking agents, such as bifunctional alkylating agents (4–14) (mitomycin C, diepoxybutane, nitrogen mustard, etc.) or furocoumarins in combination with UVA (4, 10, 14–18). In comparison to the former compounds bifunctional furocoumarins offer a number of advantages: (*a*) there is no need of metabolic activation; (*b*) treatments can be accurately controlled by the activating light; (*c*) the types of lesions induced are well defined as mono- and biadducts (interstrand cross-links) in DNA (see for review Refs. 19 and 20).

In contrast to mitomycin C (21) which reacts with purine bases, furocoumarins form only adducts with pyrimidine bases in DNA (19, 20). The latter adducts consist of pyrone-side (3,4) adducts and furan-side (4′,5′) adducts, part of which are convertible into CL. This reaction scheme offers the possibility of control of the ratio of mono- over diadducts. In fact, as shown previously by the use of different defined wavelengths in the range of 365 nm and above 380 nm, the photoreaction of bifunctional furocoumarins can be directed towards the induction of mixtures of monoadducts and DNA interstrand cross-links at different ratios or MA alone (22–25).

Up to now, much research work has been devoted to the comparison of the effects of monofunctional and bifunctional furocoumarins in order to seek an understanding of the biological role of monoadducts and DNA cross-links induced (Refs. 26–28; for review see Ref. 19). However, these attempts involved comparisons of structurally different adducts formed by different types of furocoumarins.

To avoid this bias and in order to define the contribution of MA and CL to the particular sensitivity of FA cells, we determined in the present study the lethal effects of different types of adducts formed by the same bifunctional furocoumarin, the highly photoreactive 4,5′,8-trimethylpsoralen, in combination with different wavelengths. In addition, knowing that the fate of CL in FA cells is still a matter of debate (see discussion in Ref. 18), we investigated the extent to which the sensitivity of FA cells is related to their repair capacity of both CL and cross-linkable MA (furan-side). This is a development of our previous study on CL repair in FA and normal cells after 8-MOP photoaddition (18).

MATERIALS AND METHODS

Cells and Culture Conditions. The normal human cell line used was the skin fibroblast cell line 1 BR/3, kindly provided by Dr. C. Arlett, University of Sussex, Falmer, United Kingdom. The Fanconi's anemia skin fibroblast cell lines FA 150 and FA 145 were obtained by the courtesy of Dr. R. Voss (Hadassah Hospital, Jerusalem, Israel) and Dr. M. Buchwald (Hospital of Sick Children, Toronto, Ontario, Canada), respectively. They were shown to belong to complementation groups A and B, respectively (29, 30). The passage numbers lay between 10 and 20 for all cell lines used.

Cell cultures were maintained in humidified 7% CO_2 atmosphere at 37°C using MEM (Gibco), supplemented with 2 mM glutamine, 40 μg/ml gentamicin, and 15% fetal calf serum (Boehringer, Mannheim, France S.A.).

Cell Survival. Survival experiments were performed as described previously (31). Exponentially growing cells were treated, trypsinized, resuspended in MEM, and counted in a ZM Coulter Counter (Coulter Electronics, France). Depending on the treatment dose, appropriate numbers of cells were seeded in 9–12 plastic dishes (diameter, 100 mm) (Falcon Plastics, Oxnard, CA) per experimental data point. After 16 days of incubation, the cells were fixed, stained, and counted. The cloning efficiency of control cultures was approximately 15–25% for 1 BR/3 cells and 7–12% for FA 150 cells. Surviving fractions were determined by the ratio of cloning efficiency of treated cultures (4,5′,8-trimethylpsoralen plus light) over that of control cultures incubated with 4,5′,8-trimethylpsoralen for 10 min in the dark. Survival was unaffected by treatments with 4,5′,8-trimethylpsoralen in the dark or by light treatments alone. All experiments were performed at least in triplicate.

Treatments with 4,5′,8-Trimethylpsoralen and Light. Cells in exponential growth phase were washed with PBS and incubated for 10 min in the dark in a 5 μM 4,5′,8-trimethylpsoralen solution (Sigma Chemical Co., St. Louis, MO). Cells were treated with 405 nm light alone for the induction of monoadducts or with 405 nm light, washed with 4 × 10 ml of ice cold PBS, and reirradiated with 365 nm light for the partial conversion of furan-side (4′,5′ monoadducts) into DNA interstrand cross-links. No photosensitization was observed with cells that were incubated with 4,5′,8-trimethylpsoralen in the dark, washed as above, and exposed to 365 nm light.

For DNA photobinding measurements, we used [^{3}H]-4,5′,8-trimethylpsoralen (specific activity, 12.7 Ci/mmol) purchased from HRI Associates, Inc., Emeryville, CA. Cell lysates were treated with proteinase K (Boehringer, Mannheim, Federal Republic of Germany) followed by a two step extraction with isoamyl alcohol/chloroform (1/24 vol/vol) including a treatment with RNase (10 μg/10^7 cells). After ethanol precipitation (twice), DNA was dissolved in sodium citrate buffer and the amount of [^{3}H]-4,5′,8-trimethylpsoralen bound to DNA was esti-

Received 8/3/87; revised 12/1/87; accepted 12/22/87.

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[1] This work was supported by grants from CNRS, INSERM (Contract 852017), CEA (Saclay) France, and CEE Grant BIO-151 F.

[2] The abbreviations used are: FA, Fanconi's anemia; UVA, broad band 320–380 nm radiation (peak emission 365 μm); 8-MOP, 8-methoxypsoralen; MA, monoadducts; CL, DNA interstrand cross-links; MEM, minimal essential medium; PBS, phosphate-buffered saline; MNNG, *N*-methyl-*N*′-nitro-*N*-nitrosoguanidine; XP, xeroderma pigmentosum.

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mated from the ratio of radioactivity over the amount of DNA determined spectroscopically. Contamination of DNA by low molecular weight RNA was excluded according to fluorescence measurements.

Irradiation Source. Monochromatic 365 nm and 405 nm light was obtained from a Hanovia 2.5-kW xenon lamp used in combination with a Kratos 252 high intensity grating monochromator (Kratos Analytical Instruments, Ramsey, NJ) and appropriate filters as described previously (24). The dose rate was 15–20 kJ/m^2 s at 365 and 405 nm light as determined by a YSI-Kettering Model 65A radiometer (Yellow Springs Instruments, Yellow Springs, OH).

Alkaline Elution Analysis. To determine the production and fate of 4,5′,8-trimethylpsoralen plus light induced DNA cross-links as well as furan-side (4′,5′) monoadducts, we used the alkaline elution technique worked out by Kohn *et al.* (32) with the modifications described previously (33).

Treated and untreated human cells prelabeled with 0.4 μCi/ml [*methyl*-^{14}C]thymidine (specific activity, 50.5 mCi/mmol) for 48 h which received a dose of 4 Gy of X-rays were deposited on polycarbonate filters (Nucleopore Corp., Pleasanton, CA) together with Chinese hamster V-79 cells, as internal standard, prelabeled with 0.124 μCi/ml of [*methyl*-^{3}H]thymidine (specific activity, 71.7 Ci/mmol) which received a dose of 2 Gy of X-rays. Cells were lysed for 1 h with 2% sodium dodecyl sulfate (Sigma)-0.95% tetrasodium EDTA (Sigma) at pH 9.7, washed with 0.02 M tetrasodium EDTA at pH 10 and eluted with tetrapropylammonium hydroxide (Aldrich) at pH 12.2. Residual radioactivity on the filters and in the eluted fractions was counted in a Rack Beta II (LKB, France) scintillation counter in the presence of Aquasol II (New England Nuclear). The cross-link coefficient K_c was determined according to the method of Cohen *et al.* (34):

$$K_c = \left(\frac{1 - R_0}{1 - R_1}\right)^{1/2} - 1$$

where R_0 and R_1 are the relative retentions of ^{14}C-labeled DNA from untreated and 4,5′,8-trimethylpsoralen plus UVA-treated cells, respectively, following 4 Gy of X-irradiation. Relative retention is defined as the fraction of ^{14}C-labeled human DNA retained on the filter when 50% of the ^{3}H-labeled DNA of the internal standard has eluted. In the alkaline elution profiles, an increase in elution rate corresponds to CL induction and its decrease to incision of CL (32, see also Fig. 1).

Excision of furan-side MA was measured by converting, after a period of posttreatment incubation, residual MA to CL. Cells were exposed to 405 nm light in the presence of 4,5′,8-trimethylpsoralen, washed as described above, incubated for 24 h in MEM, and thereafter reirradiated in ice cold PBS with 365 nm light. Cross-linkable furan-side MA left over after excision were determined by evaluating the extent of DNA cross-linking (as measured by the cross-linking coefficient K_c) induced by reirradiation with 365 nm light before and after the incubation period. Data shown in tables represent the means of 4 to 6 independent experiments. It should be recalled that the alkaline elution technique does not allow distinguishment of incision from complete excision of lesions since both events lead to the same changes in elution profiles. The term "incision" when used in this paper covers in fact the two events.

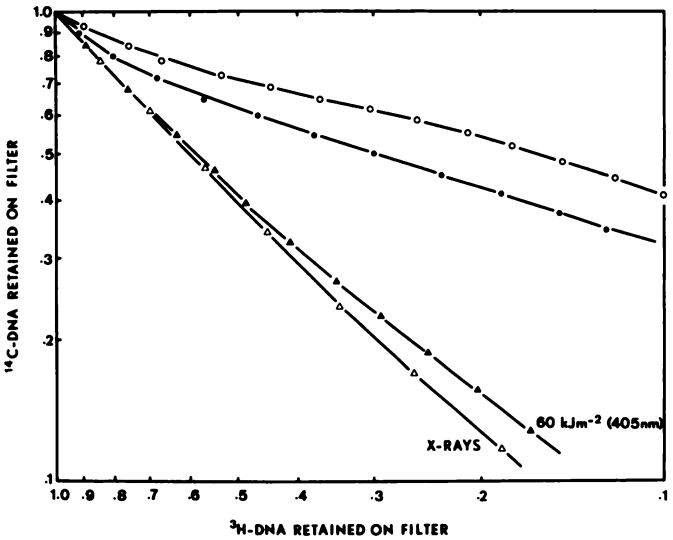

Fig. 1. Alkaline elution profiles of DNA from normal human fibroblasts 1 BR/3 treated with X-rays alone (control) (△) or with 4,5′,8-trimethylpsoralen (5 μM) and either a dose of 60 kJ/m^2 of 405 nm radiation at time 0 (▲) or a first dose of 4 kJ/m^2 of 405 nm radiation and a second dose of 1.2 kJ/m^2 of 365 nm radiation at time 0 (○) or at 24 h of posttreatment incubation (●). All samples received the same X-ray dose of 4 Gy. Data are expressed as the fractions of ^{14}C-labeled DNA retained on the filter as a function of ^{3}H-labeled DNA (internal standard).

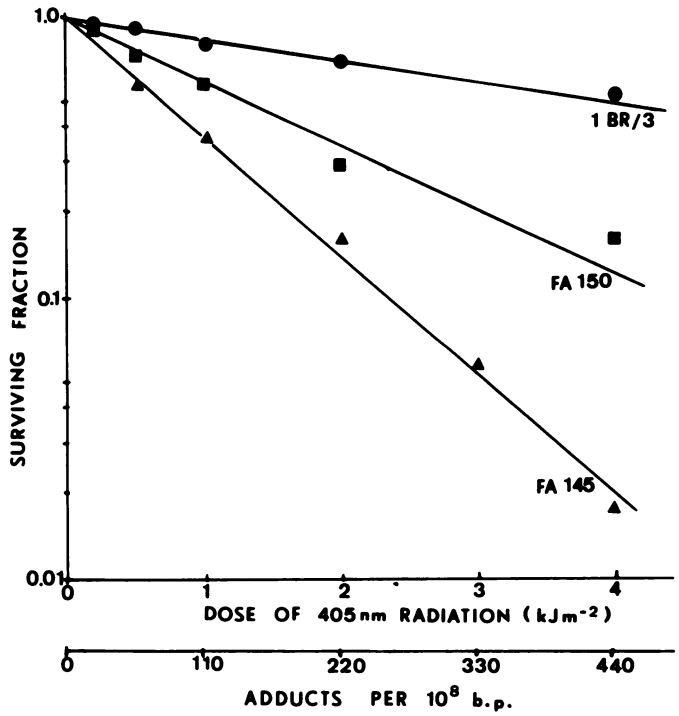

Fig. 2. Effect of monoadducts alone. Survival after photoaddition of 4,5′,8-trimethylpsoralen (5 μM) of exponentially growing normal human fibroblasts, 1 BR/3, and FA cell lines belonging to complementation group A, FA 150, and to complementation group B, FA 145, as a function of 405 nm radiation dose and adducts per 10^8 base pairs (*b.p.*).

RESULTS

Absence of Detectable Cross-Links after 4,5′,8-Trimethylpsoralen plus 405 nm Radiation at Doses Far Above the Biological Range. The possible induction of CL by 4,5′,8-trimethylpsoralen plus 405 nm radiation was checked by alkaline elution in the range of doses from 4 to 60 kJ/m^2. Results show that in the range of doses used in biological experiments and up to 24 kJ/m^2, cross-links are undetectable. It is only for a very high dose of 60 kJ/m^2 that a slight induction of CL corresponding to a K_c of 0.8×10^{-2} is seen (Fig. 1). The dose-effect relationship for CL induction at 405 nm does not follow a linear function. Consequently, it seems reasonable to assume that the role of CL, putatively induced by 4,5′,8-trimethylpsoralen and 405 nm radiation in the range of doses used in biological experiments (maximal dose, 4 kJ/m^2; see below), is negligible.

Effect of 4,5′,8-Trimethylpsoralen plus 405 nm Induced Monoadducts on Cell Survival. In view of the result described above, we used 4,5′,8-trimethylpsoralen in combination with 405 nm irradiation to determine the effect of 4,5′,8-trimethylpsoralen-induced DNA monoadducts alone. Fig. 2 shows the survival of normal and FA fibroblasts of complementation groups A and B after treatment with 4,5′,8-trimethylpsoralen as a function

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of dose of 405 nm light and number of monoadducts induced. The survival curves follow an exponential function. The results clearly demonstrate that both types of FA cells are more sensitive than normal human cells to 4,5′,8-trimethylpsoralen-induced monoadducts. FA B cells are about 5-fold more sensitive whereas group A cells are only about 3-fold more sensitive than normal human cells (1 BR/3). However, after 4,5′,8-trimethylpsoralen (data not shown) or 8-MOP and 365 nm radiation a treatment inducing a mixture of MA and CL, FA group B cells were less sensitive than FA group A cells (18).

We determined in parallel the cellular DNA-photobinding capacity of 4,5′,8-trimethylpsoralen plus 405 nm light using [^{3}H]-4,5′,8-trimethylpsoralen. Photobinding was a linear function of light dose. For the doses of 1, 2, and 4 kJ/m^2, 110, 220, and 440 adducts/10^8 base pairs were photobound to DNA, respectively (see also Ref. 35).

Effect on Survival of Increased Amounts of DNA Cross-Links Induced by 4,5′,8-Trimethylpsoralen at a Fixed Number of Monoadducts. To determine the specific contribution of DNA cross-links and of MA to the differential sensitivity of FA cells, we used the double irradiation procedure (see "Materials and Methods"): exposure to 4,5′,8-trimethylpsoralen plus 405 nm followed, after washing out of unbound molecules, by an exposure to 365 nm radiation. Under these conditions, starting with a fixed number of MA (induced at 405 nm), the amount of CL increases with the dose of 365 nm irradiation.

In order to discard the killing due to MA alone (see Fig. 2) and to assess the contribution of CL in cytotoxicity at a constant number of total MA, survival curves were normalized (see legends to Figs. 3 and 4) and expressed as a function of the second dose (365 nm).

As illustrated in Fig. 3, FA cells, belonging to complementation group A, are approximately 6 times more sensitive than normal human cells, in terms of final slope D_0, to the induction of DNA cross-links at a constant low level of total DNA adducts (20 adducts/10^8 base pairs). At a higher level of total DNA adducts (440 adducts/10^8 base pairs) induced, *i.e.*, at a higher ratio of MA over DNA cross-links, FA cells are approximately 4 times more sensitive than normal human cells (Fig. 4). This indicates that, at the same level of DNA cross-links induced (*e.g.*, a K_c equal to 1 extrapolated from data in Table 1 and Figs. 3 and 4), the differential sensitivity of FA and normal cells diminishes with increasing amounts of MA induced.

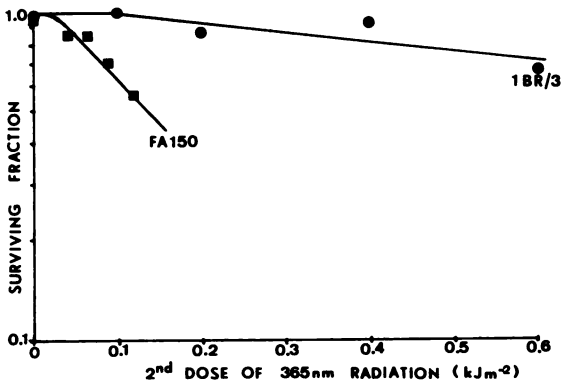

Fig. 3. Effects of cross-links at a low level of total photoadducts. Survival obtained for exponentially growing normal fibroblasts, 1 BR/3, and a FA cell line belonging to complementation group A, FA 150, after treatment with 4,5′,8-trimethylpsoralen (5 μM) and a single exposure (0.2 kJ/m^2) to 405 nm radiation followed, after washing out of unbound 4,5′,8-trimethylpsoralen, by a second exposure to 365 nm radiation. The survival curves were normalized taking the survival obtained after treatment with 0.2 kJ/m^2 of 405 nm radiation as 100%.

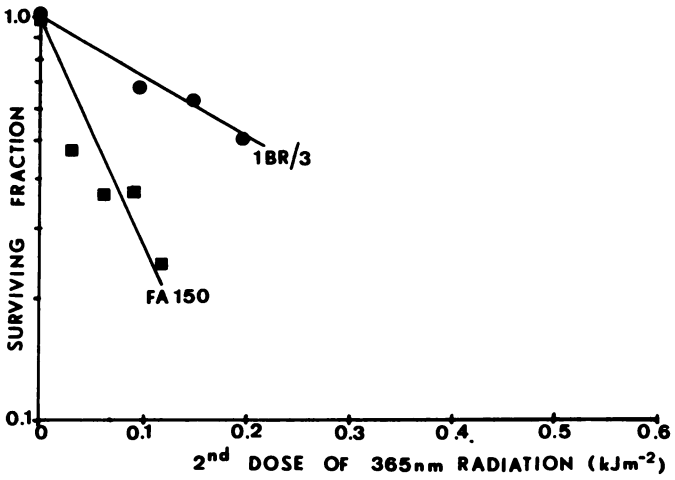

Fig. 4. Effect of cross-links at a high level of total photoadducts. Survival obtained for exponentially growing normal human fibroblasts, 1 BR/3, and a FA cell line belonging to complementation group A, FA 150, after treatment with 4,5′,8-trimethylpsoralen (5 μM) and a single exposure (4 kJ/m^2) to 405 nm radiation followed, after washing out of unbound 4,5′,8-trimethylpsoralen, by second exposures to 365 nm radiation. The survival curves were normalized taking the survival obtained after treatment with 4 kJ/m^2 of 405 nm radiation as 100%.

Table 1 *DNA cross-linking in normal, human fibroblasts (1 BR/3) induced by 4,5′,8-trimethylpsoralen and 405 nm radiation followed, after washing out of unbound 4,5′,8-trimethylpsoralen, by exposures to 365 nm radiation*

1st dose of 405 nm radiation (kJ/m^2)	2nd dose of 365 nm radiation (kJ/m^2)	Cross-linking coefficient[a] ($K_c \times 10^{-2}$)
0.2	0	0
	1.2	7.5
	2.4	15.7
	4.8	30.8
2.0	0	0
	0.8	15.9
	1.6	27.7
4.0	0	0
	0.2	9.6
	0.4	14.9
	1.2	38.7

[a] The cross-linking coefficients K_c (34) are derived from alkaline elution analysis (see "Materials and Methods").

Repair of Cross-Linkable Furan-Side Monoadducts Induced by 4,5′,8-Trimethylpsoralen and Light. Due to their absorption properties, only the psoralen-induced furan-side MA are susceptible to conversion into CL by additional exposure to 365 nm light. These MA lose their convertibility after incision or removal. This permitted us to analyze the incision of 4,5′,8-trimethylpsoralen plus 405 nm light-induced furan-side MA by their conversion into CL, after posttreatment incubation of 4,5′,8-trimethylpsoralen plus 365 nm light-treated cells. Such experiments assume that changes in chromatin structure or other variables do not affect the efficiency of CL formation. Table 2 shows the results obtained for normal human and FA cells by alkaline elution measurements. FA cells (groups A and B) are found to be less able to repair 4,5′,8-trimethylpsoralen plus 405 nm light-induced cross-linkable MA than normal human cells (1 BR/3). Twenty-four h after posttreatment incubation, FA cells from the two complementation groups incise approximately 4 times less efficiently cross-linkable furan-side MA. These results may explain, at least in part, the sensitivity of FA cells to 4,5′,8-trimethylpsoralen plus 405 nm light-induced MA (Fig. 2).

Repair of 4,5′,8-Trimethylpsoralen plus Light-induced DNA Cross-Links. As illustrated in Fig. 1, the induction and fate of

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Table 2 *Incision of 4,5′,8-trimethylpsoralen plus light-induced cross-linkable monoadducts (furan-side) and cross-links in normal human and FA fibroblasts after 24 h of posttreatment incubation*

Light dose (kJ/m²)						
405 nm (1st dose)	365 nm (2nd dose)	Total no. of adducts/ 10^8 base pairs	$K_c \times 10^{-2}$	Normal (1 BR/3)	FA 150 (group A)	FA 145 (group B)
				Incision of cross-linkable monoadducts (%)		
4.0		440	CL undetectable	40 ± 6	8 ± 5	10 ± 5
				Incision of cross-links (%)		
0.2	2.4	20	15.7	62 ± 4	47 ± 4	ND[a]
4.0	1.2	440	38.7	46 ± 3	25 ± 2	ND

[a] ND, not determined.

4,5′,8-trimethylpsoralen plus light-induced cross-links were analyzed by the alkaline elution technique. Using the 405 nm-365 nm light reirradiation protocol combining different first exposures to 405 nm light (0.2, 2, and 4 kJ/m²) with second exposures to 365 nm light, it is shown that the induction of DNA cross-links (in terms of K_c) increases linearly as a function of the second doses of 365 nm used (Table 1; Ref. 35). One example showing the induction of CL at time 0 and their decrease in normal cells after 24 h of posttreatment incubation is given in Fig. 1. The K_c values derived from similar profiles were used to estimate the percentage of repair.

In order to demonstrate incision of DNA cross-links in the presence of different levels of MA, the fate of DNA cross-links was analyzed following treatments of normal human and FA cells with either 405 nm light (0.2 kJ/m²) and 365 nm light (2.4 kJ/m²) leaving 20 adducts/10^8 base pairs, or with 405 nm light (4 kJ/m²) and 365 nm light (1.2 kJ/m²) leaving 440 adducts/10^8 base pairs. As seen in Table 2, in the two conditions, at equal numbers of total adducts and CL, FA cells incise 1.3 and 1.8 times less than normal cells, respectively. In other words, the difference in CL incision between normal and FA cells becomes more pronounced when increasing the amount of MA by a factor of 20 and that of CL by a factor of 2.4. Additional experiments have shown that in normal as well as in FA cells a 5-fold increase in cross-linking level (K_c) does not change the efficiency of CL incision (data not shown). Thus, MA appear to be principally responsible for the decline in CL incision in both cell lines. Moreover, since the decrease in CL incision capacity between the two conditions is more pronounced in FA than in normal cells (Table 2), it can be concluded that CL incision is more affected by the presence of MA in FA than in normal cells.

DISCUSSION

It has been recently shown that bifunctional furocoumarins in combination with irradiation at specific wavelengths are able to induce selectively only MA at wavelengths above 380 nm or mixtures of mono- and biadducts at wavelengths around 365 nm (22–25). We indeed confirmed that CL are undetectable in cellular DNA after treatment with 4,5′,8-trimethylpsoralen and 405 nm radiation for doses far above the range used in biological experiments. Furan-side MA, in suitable sequences in DNA, can be converted, after washing out of unbound furocoumarin molecules, into CL by a second exposure to 365 nm radiation (20). Such conversion cannot take place when furan-side MA are incised or removed during cell incubation between the 405 nm and 365 nm irradiations. The actual work is based on these premises. FA cells are usually considered to be only marginally susceptible to cytotoxicity due to DNA damage induced by UV (36), ionizing radiation (37, 38), and monofunctional agents, such as decarbamoyl mitomycin C, methyl methanesulfonate, or 4-nitroquinoline oxide (4, 9, 10, 15, 39). However, it is of interest to note that two FA lymphoblast cell lines (FA 62 and FA 230) now known to belong to complementation group B and one cell line (FA 72) belonging to group A were found to be clearly sensitive to the monofunctional alkylating agent MNNG (11).

Here we demonstrate that FA cells are not only more sensitive than normal cells to 4,5′,8-trimethylpsoralen treatments producing a mixture of CL and MA as already established for 8-MOP (18) but they are also more sensitive to 4,5′,8-trimethylpsoralen-induced MA (Fig. 2). To our surprise, FA group B cells were found to be even more sensitive than group A cells, whereas the reverse was observed for treatments leading to a mixture of MA and CL, *i.e.*, produced by 8-MOP. More experiments are required to explain this difference (other cell lines, dose-response for 4,5′,8-trimethylpsoralen plus 365 nm, etc.). The differential sensitivity of two FA cell lines as compared to normal cells to MA alone is in line with the observation that the efficiency of repair of furan-side MA is decreased in both FA cells (Table 2). As we discussed at length in Ref. 18, various authors have obtained conflicting data with respect to CL incision in FA compared to normal cells. We can recall that following 8-MOP photoaddition (365 nm) we observed a correlation between cellular sensitivity and CL reparability (group A < group B < normal) (18). In other words, FA cells belonging to complementation group A appear to be more defective than group B cells for the incision of CL, whereas group B cells seem as defective as group A cells in incision of MA at least at 24 h following treatment. It should be stressed that FA cells are impaired in both MA and CL incision (Table 2) compared to normal cells.

It is generally believed that incision of CL and MA are under the same genetic control. This is based on the fact that XP(A) and XP(D) human cells (26, 40), as well as *rad1, rad2, rad4* excision-defective yeast mutants (41) are deficient in incision of both MA and CL induced by photoaddition of psoralens. However, it has been reported recently (40) that an XP revertant from group A is capable of removing CL but not MA. The reverse situation has been found for the *rad3* yeast mutant which is blocked in the incision of CL (41–43) but is proficient in MA incision (41). Taken together with our data on the differential responses in groups A and B of FA cells, these observations suggest that subtle differences may exist in the recognition and/or incision step(s) for each type of lesion, *i.e.*, MA and CL.

Some of the properties described for FA cells recall those reported for certain repair deficient mutants in yeast. Similarly to FA group A and B cells, mutant *pso1-1* and *pso2-1* cells were

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found to be defective in the repair of furocoumarin plus light-induced lesions (44). *pso2-1* mutant cells were very sensitive to the induction of CL and fairly insensitive to furocoumarin-induced MA, whereas *pso1-1* mutant cells were sensitive to the induction of both DNA CL and MA (45, 46).

As shown in the present paper with 4,5′,8-trimethylpsoralen, the reirradiation method combining a first exposure to 405 nm with a second exposure to 365 nm light permits determination of the effects of increasing levels of DNA cross-links at a chosen number of total photoadducts. The data obtained on cell survival extend previous results obtained with the bifunctional furocoumarin 8-MOP (18) in providing unambiguous evidence for an increase in differential sensitivity of FA cells (group A) *versus* normal cells due to the presence of CL. Moreover, we demonstrate here that, when decreasing the number of MA from 440 to 20 adducts/10^8 base pairs, in the presence of CL, the differential sensitivity between FA group A and normal cells increases from 4- to 6-fold (Figs. 3 and 4). From this, it can be concluded that the sensitivity of the cells (FA group A) to 4,5′,8-trimethylpsoralen-induced MA masks in part the sensitivity to 4,5′,8-trimethylpsoralen-induced DNA cross-links.

Alkaline elution analysis of DNA from FA and normal human fibroblasts treated with 8-MOP and 365 nm light showed already that FA group A cells possess approximately one-half of the capacity of normal cells to incise DNA cross-links during 6 h of posttreatment incubation (18). Using the 405 nm–365nm reirradiation protocol with the bifunctional furocoumarin 4,5′,8-trimethylpsoralen, we show here that for a constant number of total adducts or different ratios of CL over MA, the repair of CL is systematically hampered in FA (group A) compared to normal cells (Table 2).

The excision repair pathway is assumed to be active against DNA adducts that cause major helical distortions. It has been recently shown that the bacterial *uvr* ABC excinuclease participates in the repair of DNA damage induced by MNNG (47). This is interpreted as indicating that besides the lesions repaired by the methyl transferase and glycosylases, MNNG produces lesions which lead to considerable helical distortions. The MA induced by 4,5′,8-trimethylpsoralen plus 405 nm light are also likely to produce perturbations in the stability of the double helix. Since FA cells turn out to be sensitive to both MNNG and 4,5′,8-trimethylpsoralen plus 405 nm light, whereas they are only slightly sensitive to other monofunctional agents tested and to radiations, it can be proposed that up to a certain minor qualitative and/or quantitative level of helical perturbations, the incision complex in FA cells will act as in normal ones; the DNA adduct itself is either not recognized by the excision complex or processed in both cell types with the same efficiency. Even a fraction of the CL which are indeed incised in FA cells (Table 2; Ref. 18) may fall in this last case. It would be only at high levels of helical perturbations that the incision complex in FA cells would manifest its defect. In other words, it appears that with respect to the relative sensitivity of FA cells to genotoxic agents, it is not the mono- or bifunctional nature of the DNA adducts which matters but the degree of helical perturbation induced.

ACKNOWLEDGMENTS

The authors wish to thank C. Vivès and D. Chardonnieras for excellent assistance.

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